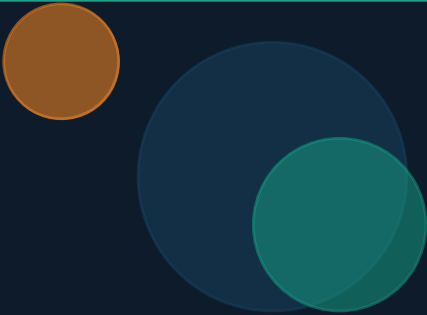




# Arduino-Based RFID Attendance System

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# Aim & Objectives

## AIM

To design and implement an automated, reliable RFID-based attendance system using Arduino Uno that records students presence in real-time, logs data to a CSV file on an SD card for Excel analysis, and provides immediate audio-visual feedback.

01

### Automated Identification

Scan ID tags to identify registered users without manual entry.

02

### Real-Time Timestamps

Use DS1302 RTC module to record exact date and time of each attendance event.

03

### Persistent CSV Logging

Append records to ATTEND.csv on SD card, readable directly in Microsoft Excel.

04

### Instant Feedback

Provide LED +buzzer signals and LCD display of user name and timestamp on each scan.

05

### Error Handling

Detect and report SD card failures or invalid scan events on the 16x2 LCD.

# Hardware Components

## Arduino Uno

ATmega328P Microcontroller  
Main processing unit

## MFRC522

RFID Reader Module  
13.56 MHz, SPI interface

## SD Card Module

SPI Storage  
CSV data logging

## DS1302 MH Real Time Clock

Real-Time Clock  
Date & time stamping

## 16×2 LCD + I2C

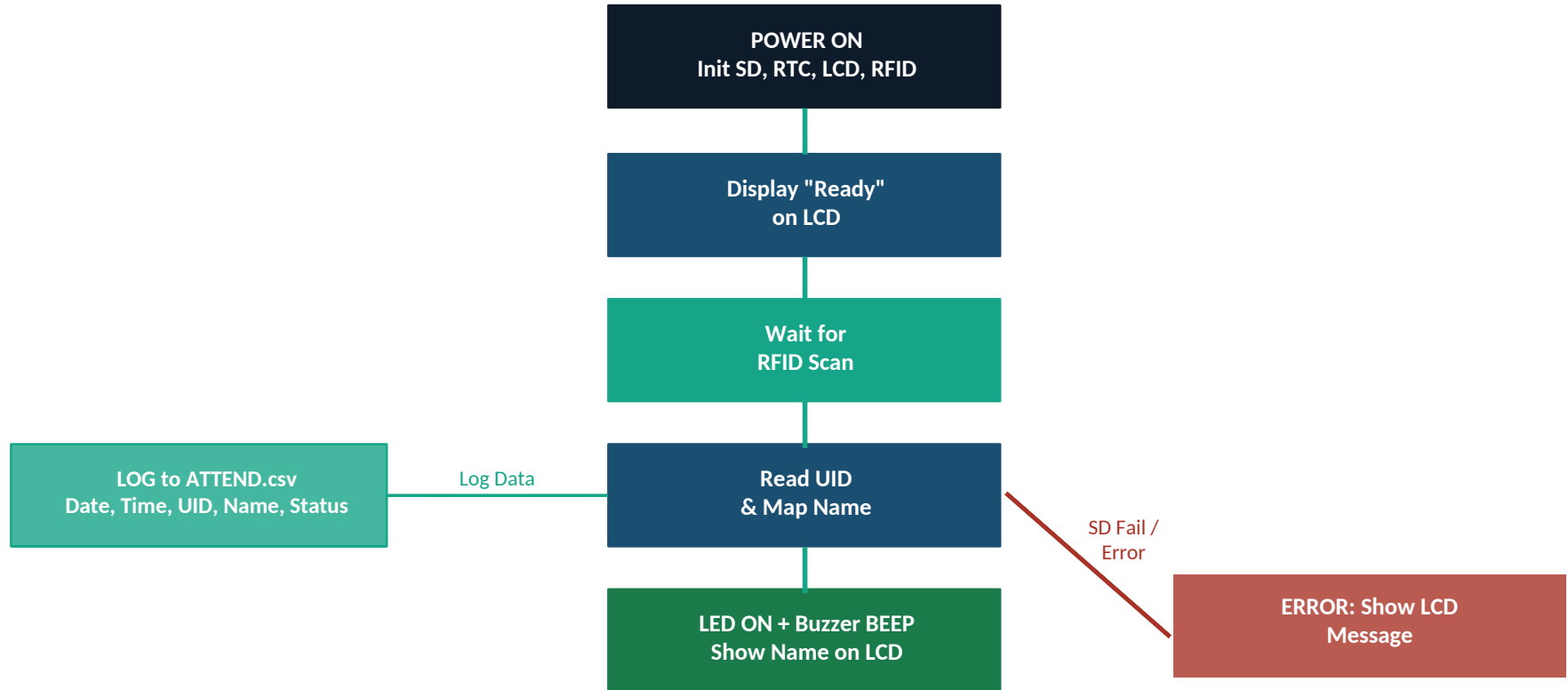
Display Module  
User feedback screen

## LED + Buzzer

Active indicators  
Visual & audio feedback

Communication: RFID + SD Card via SPI | RTC + LCD via I2C | LED + Buzzer via Digital Pins

# Workflow & System Architecture



# Implementation Details

## Code Logic (Arduino)

//Libraries Used:

```
#include <SPI.h>
#include <MFRC522.h>
#include <SD.h>
#include <RTCLib.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
```

// On RFID scan:

```
String uid = getUID();
String name = lookupName(uid);
DateTime now = rtc.now();
File f = SD.open("ATTEND.csv",
FILE_WRITE);

f.print(date + "," + time + ","
+ uid + ","
+ name + ",Present");
lcd.print(name);
digitalWrite(LED, HIGH);
tone(BUZZER, 1000, 200);
```

## UID Mapping

Registered UIDs stored in arrays; matched on scan to return user name and status.

## CSV Format

ATTEND.csv with headers: Date, Time, UID, Name, Status – fully Excel-compatible.

## RTC Integration

DS1302 provides accurate timestamps ( $\pm 2$ ppm). DateTime object formats date/time strings.

## SPI Bus Sharing

RFID(SS:pin10)and SD Card (SS: pin 4) share MISO/MOSI/SCK with separate chip-selects.

## LCD I2C

16×2display at address 0x27 shows user name + timestamp + system status messages.

# Data Visualization

|    |            |          |             |                    |                  |
|----|------------|----------|-------------|--------------------|------------------|
| A1 |            |          |             |                    | Date             |
|    | A          | B        | C           | D                  | E                |
| 1  | Date       | Time     | UID Decimal | Student Name       | Attendance Taken |
| 2  | 17.05.2026 | 23:28:17 | 369715717   | Nurbol Makhmudulla | YES              |
| 3  | 17.05.2026 | 23:28:20 | 2386988548  | Tom Crusie         | NO               |
| 4  | 17.05.2026 | 23:28:23 | 3546464770  | Bekzat Tokbolat    | YES              |
| 5  | 17.05.2026 | 23:28:26 | 68361902    | Abdinur Otegen     | YES              |
| 6  |            |          |             |                    |                  |
| 7  |            |          |             |                    |                  |
| 8  |            |          |             |                    |                  |
| 9  |            |          |             |                    |                  |
| 10 |            |          |             |                    |                  |
| 11 |            |          |             |                    |                  |

# System Demonstration

LCD 20x4 — Successful Scan

=====

Welcome!!

User ID: 1

=====

LCD — Error State

=====

Access Denied

Unknown Card

=====

## Live Demo Steps

1

Power On

2

Scan RFIDTag

3

Checking

4

Display LCD, Buzzer, LED or Error

5

Excel Data

# Conclusions

## Automated Attendance

Fully automated RFID-based attendance with zero manual data entry.

## Accurate Timestamps

DS1302 provides precise, battery-backed date/time for each scan.

## Excel-Ready CSV Logging

ATTEND.csv on SD card is directly importable into Microsoft Excel for analysis.

## Instant Feedback

LED+buzzer+16×2LCD give clear confirmation on every successful scan.

## Robust Error Handling

ID card failure and scan errors are gracefully reported via LCD.